

Applied A.I. Solutions

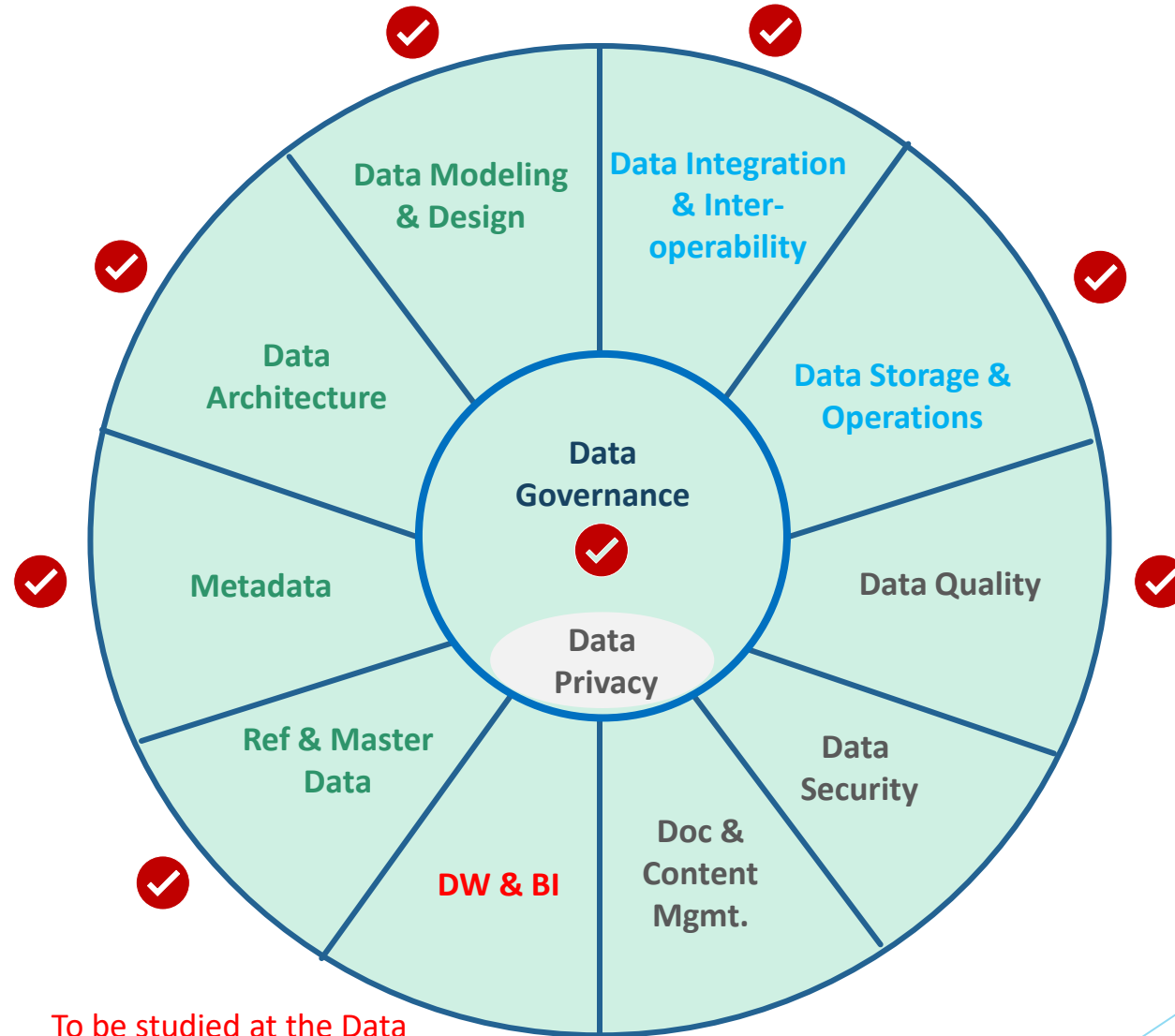
Foundations of Data Management

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DATA QUALITY

The DAMA Wheel



To be studied at the Data Visualization Techniques course

INTRODUCTION

- DM involves a set of complex, **interrelated processes** that enable an organization to use its data to achieve strategic goals
- **DQ within the DMF includes managing data through its entire lifecycle**
- DQ programs have data-related operational **responsibilities** and **accountabilities**

Definition

The planning, implementation, and control of **activities that apply quality management techniques to data**, to assure it is **fit for consumption** and meets the needs of data consumers

DQ programs are guided by the following **principles**:

- **Criticality**
- Lifecycle Management
- **Prevention**
- Root cause remediation
- Governance
- Standards-driven
- Objective measurement
- **Transparency**
- **Embedded in business processes**
- Connected to **service levels**

Drivers

- **Increase** the **value** of organizational data
- **Reduce risks** and costs associated with poor quality data
- Improve organizational **efficiency** and productivity
- Protect and enhance the organization's **reputation**

Goals

- Develop a **governed approach** to meet DQ requirements
- Define **standards**, processes, metrics for **DQ Monitoring and Control**
- Identify, **advocate** for opportunities to improve data quality



Business Drivers

Inputs

- Data Policies
- Business rules
- Data requirements
- **Metadata**
- Data lineage



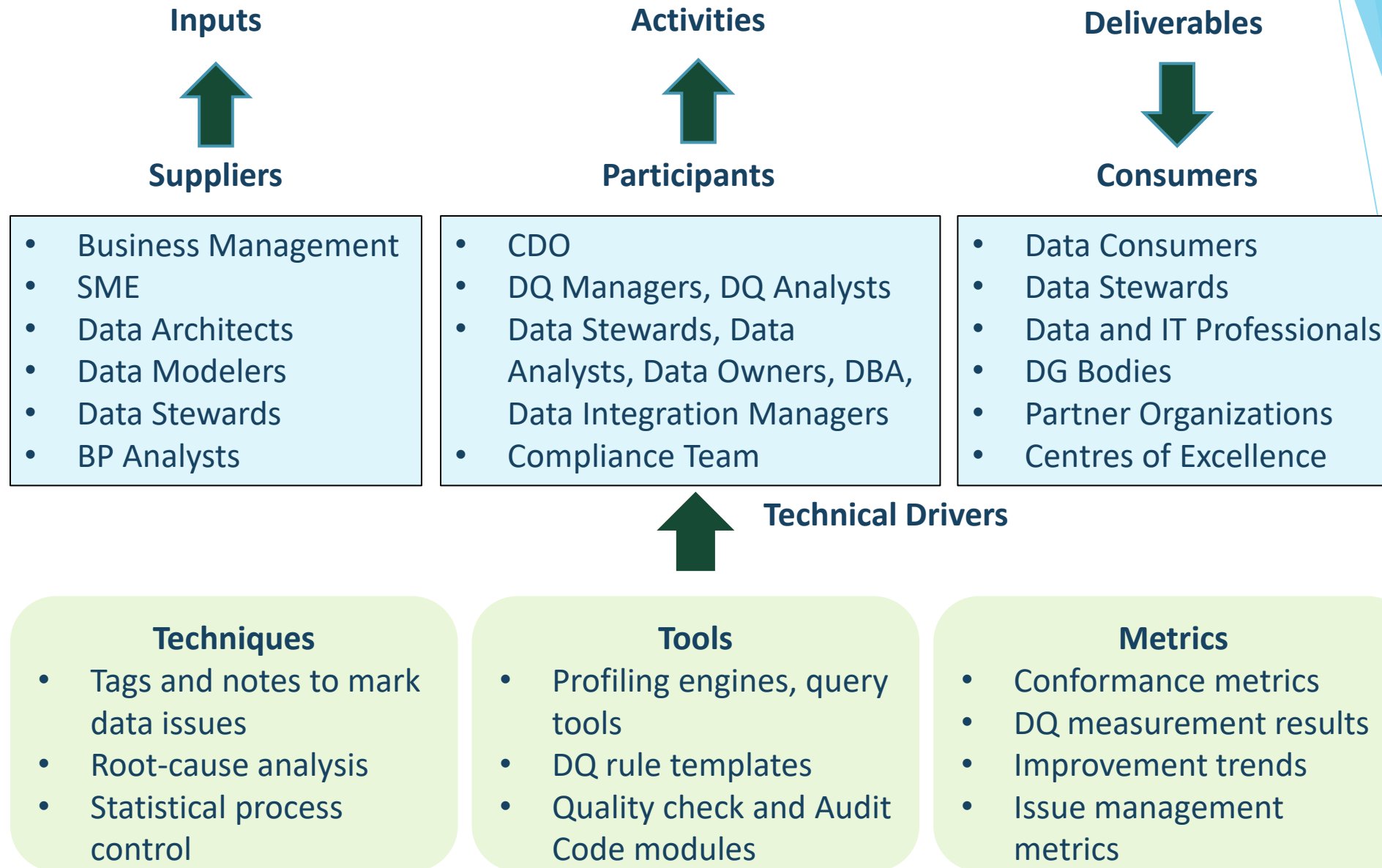
Activities

1. Define a DQ strategy
2. Define high-quality data
3. Perform DQ assessment
4. Identify and prioritize improvements



Deliverables

- **DQ Strategy**
- DQ Governance reports
- DQ Policies, Guidelines, SLA
- DQM procedure reports

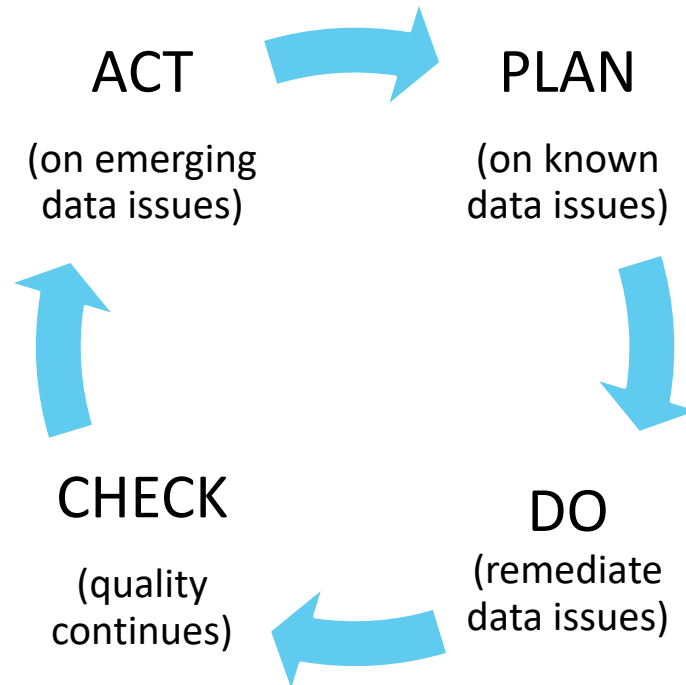


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CONCEPTS

- **Data Quality Dimensions - ICRA**
 - **Intrinsic** – accuracy, objectivity, believability, reputation (trust)
 - **Contextual** – value-added, relevancy, timelines, completeness (metadata)
 - **Representational** – easy to understanding, consistent, concise (consistency)
 - **Accessibility** – accessibility, and access security (availability)

- **Data Quality Improvement Lifecycle** (Shewhart / Deming cycle)



Problem solving model

1. Condition of data must be measured against standards
2. If it does not meet standards, root-cause(s) must be identified and remediated
3. Once remediated, data should be monitored to ensure that it continues to meet requirements and hence quality of data

- **DQ Business Rules:**

- Definitional conformance
- **Record completeness**
- Format compliance
- Value domain membership
- Range conformance
- Mapping conformance
- Consistency rules
- Accuracy verification
- **Uniqueness verification**
- Timeliness validation

- **Data Profiling to ensure DQ**
 - **Data inspection** and data quality assessment
 - Statistical techniques use to **identify** patterns, rules, constraint validity
 - **Identify** possible duplicates, missing keys, faulty referential integrity
 - **Analyze** conformity to rules and requirements
 - **Isolate** the root causes of issues

- **Data Processing to improve Data Quality**
 - Data Cleansing
 - Data Enhancement
 - Data parsing and Formatting
 - Data Transformation and Standardization

- **Data Correction**

- **Preventive Actions**

- ✓ Data entry controls
- ✓ Training
- ✓ DQ Rules
- ✓ DQ SLA

- **Corrective Actions**

- ✓ Automated correction based on rule-based standardization and normalization
- ✓ Manual correction

- **Data Governance - Quality Measurement**

Measure and Monitor Data Quality at

- **Level of detail related** to the execution of individual rules
- **Level of aggregated data** from the rules

$$\text{Valid Data (r)} = \frac{\text{Test Executions (r)} - \text{Exceptions Found (r)}}{\text{Test Executions (r)}}$$

$$\text{Invalid Data I (r)} = \frac{\text{Exceptions Found (r)}}{\text{Test Executions (r)}}$$

Dimensions & Business Rule	Metrics	Status Indicator (threshold)
Completeness	# recs populated/ total # recs * 100	Unacceptable <80% populated
Uniqueness	Duplicate # recs/ total # recs * 100	Unacceptable > 0%
Timeliness	Incomplete trans/ total # trans in a time period * 100	Unacceptable <99% completed on time
Validity	# recs condition met/total # recs * 100	Unacceptable <99% met

